Rel-SA: Alzheimer's Disease Detection using Relevance-Augmented Self-Attention

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Abstract: Alzheimer's Disease (AD) is a progressive disorder that forms lesions in the brain. Unfortunately, the harm this condition does to the brain is irreversible, which is why the importance of early detection ought to be obvious. Conventionally, the diagnosis of neurodegenerative diseases through MRI scans has been done through either visually defined brain atlases or deep learning modeling. However, both of these methods often take inordinate amounts of time to compute, and the results are all but random. This paper tackles the development of a rapid explainable and accurate detection of Alzheimer's Disease through the use of D MRI scans. This is done through a modification of the ResNet 18 coupled with Grad-CAM and the Rel Attention mechanism architect in three phases. First, a weak ResNet is trained to pinpoint MR classifications into 4 classes of dementia, namely, Non-Demented, Very Mild, Mild, and Moderate, all while generating Explainable AI and determining class specific brain mapped regions and using these to formulate a subtractive base to be used later in the combined neuron extraction. Thus, the main surpassing efforts of this paper is the 2D MI of Alzheimer's Disease Staging, Relational moderation, and hybrid convolution with Attention research to surpassing. This development of a neural base to Atlas-Def central guided features is what we are addressing with relevance-guided attention for improved transparency and diagnostic value.

Keywords: Rel_SA, Alzheimer's Detections, Atlas, ROI

I. INTRODUCTION

The goal for the progressive neurodegenerative disorder represented by Alzheimer's disease include structural and functional changes in the brain. In most cases, especially affecting the hippocampus among other regions, MRI allows for the detection of atrophy patterns during early stages. This work proposes an explainable AI-based system relying on CNNs, Explainable AI heatmaps, and a Relevance-Augmented Attention mechanism to provide transparent diagnosis of Alzheimer's Disease.

This report presents the study of the method proposed in the paper "Rel-SA: Alzheimer's Disease Detection using Relevance-augmented Self Attention by Inducing Domain Priors in Vision Transformers." The present work investigates the challenge of detecting Alzheimer's Disease (AD) from 3D structural MRI, and emphasizes that deep learning models should be interpretable and neuro scientifically grounded. This paper underlines that classic CNNs and Transformer-based architectures often do not include meaningful biomarkers from a clinical perspective. Rel-SA embeds domain-specific priors derived from brain atlases directly in the attention mechanism of the Vision Transformer. The goal of this project is to replicate all methods on 2D structural MRI scans, reproduce the reported results, and explore some possible improvements. While the base paper generates relevance maps based on fixed brain regions, the current method produces Explainable AI heatmaps that highlight Alzheimer's areas directly for more intuitive and visually interpretable explanations of the models. These new

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methods may help the model find smaller or even deeper details in MRI images that are helpful for diagnosis.

II. Background

Computer aided diagnosis in MRI tries to capture the patterns of atrophy at the beginning stages; however the traditional deep learning approaches struggle. For 3D MRI models, there is a heavy reliance on atlas-based segmentation, where professionals delineate the anatomical regions of interest (ROIs) manually. These atlases are very precise and detailed; however, they can also be problematic. These atlases are rigid and will suppress the ability to capture the patient-specific subtile variations and the other more granular pathological factors. Also, 3D CNNs need a lot of processing power which restricts the ability to be used in everyday clinical practice. The recent work on the transformer-based models, Vision Transformers (ViT) in particular, show a lot of promise in various imaging tasks. However, in these models, every spatial token is treated uniformly which indicates that there are no inductive biases about the clinically relevant structures in the brain. This is problematic as these models would pay attention to non-pertinent regions of the image which would inhibit the accuracy and interpretability of the work in question. Following the ideas set with Rel-SA, the current work is also more streamlined into a 2D pipeline. Using atlas-based regions is no longer a requisite. This is because we used Explainable AI on a CNN that is fine-tuned and relevant maps were created that box pathological regions from the training data, eliminating the need for segmentation drawn out by experts.

A. *REL – SA ARCHITECTURE FOR 2D MRI ALZHEIMER'S DISEASE DETECTION*

As part of a model, we carry out a series of preprocessing feats on the structural MRIs, the 2D grayscale ones, such as Z-score normalisation, bias field correction, and skull stripping to achieve clean and uniform entry points. Following the over lapping variant of the global patch sampling decay-schedule conduct, we snap the images into right-angled, non-art overlapping sybil 16-headed and 32-headed tiled patterns, after which, each bordered section of the image field is preferentially placed on a neural patch-embedding higher dimensioned point of neural attention and flattened spacial metrics. At the very core of the model stands the Rel-augmented Self-Attention module which, quilting with a domain bias, or a patch of a chosen region of focus, of the white matter of the AAL3 and JHU, atlases and spends valuable computational focus on the pruned-vision more salient brain regions. The attention then passes with bias, sampling from each of the layers of a transformers encoder as data passes into them serially refraining from inner-attractor synthesized feedback to a computationally stable mean pooled attention-hotspot region-averaged focus gated memory. At the final layer, a point neural junction of size one is applied as the hidden layer, with the emergent data being split into two classes, either the one corresponding to patients with Alzheimer's Disease or the one corresponding to control patients.

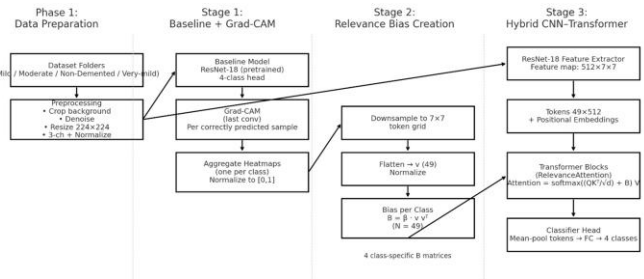
B. *Rel-SA: Relevance-Augmented Self-Attention*

The base paper describes Rel-SA as being a technically straightforward approach to assimilating the knowledge garnered from neuroscience atlases into the ViT self-attention module. More precisely, we implement an attention matrix bias relative to the softmax, where the model is nudged to focus its attention to areas of the brain that are important to the diagnosis of Alzheimer's disease. Formally, we implement the attention score as Rel-SA. In Rel-SA, clinically relevant information is incorporated within the attention mechanism bypassing the relevance bias (Rel-Bias) which corresponds to the brain regions of interest during the Alzheimer's Disease diagnostic process. This is as per the AAL3 and JHU WM brain atlases which highlight regions of the brain that are associated with Alzheimer's Disease. The attention matrix is augmented with the relevance bias before softmax to focus the attention on the clinically significant regions of interest. This can be done by masking and weighting brain regions that are known to be significant for the progression of Alzheimer's disease (AD) such as the hippocampus and amygdala, during the softmax process.

A stack of relevance-biased patches is inputted into the transformer encoder along with their tokenised sequence. The encoder then performs relevance embedding by way of the transformer blocks. Each of

these blocks is comprised of a multi-head self-attention layer, multiple feedforward and normalization layers, and a layer of layer normalization. The frozen transformer blocks have their output embeddings pooled in some way, like global average pooling, or by selecting a specific token, and then that pooled embedding is sent to a fully connected layer which is a standard classification MLP. Typically, this is binary for the detection of Alzheimer's Disease, separating it from CN (Cognitive Normal).

III. MODEL ARCHITECTURE



A **relevance bias matrix** is like a smart "guidance map" that tells the model where it should look more carefully in a brain MRI image. Instead of treating every part of the brain slice equally, the matrix highlights the areas that are most important for detecting Alzheimer's or different dementia stages. These important areas may include regions like the hippocampus, temporal lobes, or other structures that doctors know tend to show early signs of damage.

Building on the success of the ScoreCAM dual-branch model, the primary future direction is to validate its generalizability by applying it to more complex and diverse medical imaging datasets. This involves extending the methodology beyond 2D images to analyze 3D volumetric data like CT and MRI scans, which present unique spatial challenges. Furthermore, testing the model on entirely different diagnostic domains, such as dermatology for skin cancer or ophthalmology for retinal diseases, will be crucial. Proving the architecture's effectiveness across varied data types and medical tasks will demonstrate that the underlying principle is a fundamental advancement in medical AI, not just a solution for a specific problem.

Next, research should focus on evolving the model's architecture by exploring more sophisticated integrations of attention mechanisms. Moving beyond the current dual-branch design, future work could investigate deeply integrated systems where the attention map and the base CNN interact dynamically, perhaps using Transformer-based fusion for a more nuanced understanding. In parallel, it is essential to integrate emerging Explainable AI (XAI) techniques. By combining ScoreCAM's localization strengths with newer methods like concept-based models, a hybrid system could provide a richer explanation, showing not only *where* the model is looking but also *what* clinical concepts it has identified, thereby creating a more comprehensive and trustworthy AI tool.

Ultimately, the most critical step is to bridge the gap between technical performance and clinical impact through rigorous human-in-the-loop studies. This involves designing controlled experiments where clinicians diagnose cases with and without the AI's assistance, specifically evaluating whether the ScoreCAM saliency maps improve diagnostic accuracy, speed, and confidence. Such studies are necessary to validate the core promise of interpretability that the model's explanations are genuinely helpful and trustworthy to its intended users. Successfully demonstrating this clinical utility is the final and most important milestone in transitioning this research from a promising prototype to a practical, life-saving application in medicine.

A. BASELINE CNN MODEL AND EXPLAINABLE AI

A.1 GRAD-CAM

A ResNet-18 model, which has been pretrained, is being adjusted for four classifications for dementia which are Non-Demented, Very Mild Dementia, Mild Dementia, and Moderate Dementia for the first classifier in the study. This study used ResNet-18 in this model because it has the most representational power in a computationally efficient way. This is why it is most suited for 2D MRI slice classification. In the model, during the training, the model learns the most discriminative spatial pattern which is

ultimately used for the progression of the disease. This is needed because the disease has patterns such as cortical thinning, ventricular enlargement, and hippocampal atrophy. After the model finishes training, the samples that are classified correctly are used in a technique called Explainable AI (Gradient-weighted Class Activation Mapping) to map the areas that are most influential to each prediction.

Gradient activation class mapping takes the class score that is predicted, and is used to make a heat map that is a visual representation of the areas in the slice of the MRI that the model believes is the most important for the prediction. This is used to extract the data that is needed to make a pathological signature that is specific to a class. For the different dementia stages, hundreds of Explainable AI maps are made which can then be used to produce a single representation of the data for each class. For the purposes of this representation, it is able to reduce noise, improve the visibility of specific areas that are consistently associated with the disease, and show specific data-driven changes related to the anatomy that are tied to the different stages of the disease.

A.2 LAYER CAM

The baseline model employs a modified and fine-tuned version of ResNet-18 that has been pre-trained to accurately classify dementia into four distinct classes and to distinguish relevant MRI features including hippocampal atrophy and cortical thinning. To explain the models predictions, Layer-CAM is used to obtain high resolution diseased-relevant saliency maps. Layer-CAM differs from Grad-CAM which uses the last convolutional layer, Layer-CAM uses positive gradients from other intermediate layers which results in sharper, more localized activation maps that are better at capturing region of interest. For each of the correctly classified samples, Layer-CAM generated heatmaps which are then combined to obtain class-specific Relevance Priors that highlight common structures in different dementia classes. These Relevance Priors significantly improve spatial resolution and help other model units to focus on diseased-relevant structures while retaining explainability.

A.3 SCORE CAM

The initial classification framework uses a Broadway pre-trained ResNet-18 model fine-tuned on four different classes of dementia to obtain spatial features related to the Alzheimer's Disease progression. For model interpretability, relevance maps are created using Score-CAM, which highlights the most important areas for model predictions. In contrast to Grad-CAM which uses backpropagation, Score-CAM derives saliency maps by observing changes in the confidence level for a particular classification when the input image is masked with different individual activation maps. This technique of explanation is gradient-free and therefore, more stable and less noisy which is necessary in a domain such as medical imaging. For every correctly predicted MRI scan, heatmaps using Score-CAM are created such that their output is combined to yield priors of individual classes that reflect structured dementia stage changes. Based on activation maps, these priors constructed from the observation of confidence are clinically interpretable, signaling to the attention layers, and also preserving model traceability.

B. RELEVANCE-ATTENTION MODULE

The core development in which the proposed architecture stems from is the Relevance-Augmented Attention module. This module allows the Transformer to selectively attend to clinically relevant regions in the brain rather than paying attention in a spread-out manner to the entire MRI slice. This module is inspired by the Rel-SA framework. This module uses class- Explainable AI maps as a relevance external attention bias which steers the model towards pattern recognition in anatomy that is related to Alzheimer disease. Attention is focused on high-impact regions and are integrated from the baseline CNN on the hippocampus, medial temporal lobes, and cortical structure to which dementia-atrophy is severity progressive.

To utilize the maps in the attention mechanism, the accumulated Explainable AI maps are aligned in space to the tokenized feature grid from ResNet-18.

Each token of size 7 x 7 represents a patch of the MRI slice and the relevance values corresponding to the patch are set into a bias matrix that alters the attention score calculation in self-attention. In a simple Transformer model, the attention scores are calculated using a dot product between the query and the key vectors. In the Rel-Attention module, an added bias term is been derived from the specific class prior to the relevance and is injected to the calculation, which effectively increases the attention weight that is clinically important that surppresses contributions that are from unnecessary no9sy regions.

C. *HYBRID CNN–TRANSFORMER ARCHITECTURE*

Combining the best of convolutional neural networks and Transformer-based global attention, the hybrid model constructs a strong and interpretable diagnostic model. The first portion of the design is a feature extractor module. This module is a pretrained ResNet-18. After processing each input MRI slice, ResNet-18 outputs a 512, 7, and 7 feature tensors, where 512 represents the number of channels that the model is using to store the high-level spatial representation associated to the varying structures of the brain at the different stages of dementia. That tensor is then flattened to form 49 spatial tokens corresponding to distinct regions of the brain. Some positional embeddings are added to retain lost spatial relationships during tokenization, allowing the Transformer to reason contextually about the brain's structures. tokens are processed to a stack of the Transformer encoder layers where the mechanism of relevance-biased attention proposed in this research is particularly valuable.

Later different rounds of attention and transformation, a mean pooling operation is applied through all tokens to generate a compact representation of the whole MRI slice. At last, a fully classification head maps this representation to one of the four dementia classes.

IV. PAPER COMPARISION

A. *RELEVANCE SOURCE*

Rel-SA: Uses predefined, static brain regions from AAL3 and JHU atlases. These priors capture general disease hotspots but fail to adapt to individual patient variations or other datasets.
Our Approach: Employs Explainable AI based heatmaps generated from a pretrained ResNet-18 baseline model. Instead of relying on experts, your system learns relevance directly from the data, automatically identifying regions that the CNN deems diagnostically important.
Advantage:
No manual region annotation or atlas registration required. Dynamic and dataset-adaptive can generalize to new diseases (e.g., Parkinson's, tumor classification). Reduces preprocessing complexity and human dependency.

B. *CLASSIFICATION TASK*

Rel-SA: Binary Alzheimer's detection (AD vs CN), focusing purely on distinguishing healthy vs diseased.
Our Approach: Extends to **four severity stages** (Non-Demented, Very Mild, Mild, Moderate). This is more **clinically actionable**, since neurodegeneration occurs along a continuum.
Advantage: Enables monitoring disease progression. Facilitates earlier interventions. Improves real-world relevance of the diagnosis.

C. *RELEVANCE ADAPTATION*

Rel-SA: Applies a **single relevance bias** for all samples a one-size-fits-all attention prior.

Our Approach: Introduces **class-specific relevance matrices**, producing distinct bias patterns for each dementia stage. These biases are computed from class-aggregated *Explainable AI* maps and guide the Transformer to focus on **stage-relevant structural features**.

Advantage: Captures the **evolving neuroanatomical signatures** of dementia. Allows personalized interpretation of progression patterns. Strengthens feature-attention alignment across the model.

D. INPUT MODALITY

Rel-SA: Operates on 3D MRI volumes →high VRAM and long inference times.

Your Approach: Uses **2D MRI slices**, resized to 224×224, normalized, and processed as independent instances.

Advantage:

Reduces computational cost . Works with **standard hospital MRI data** where 3D alignment is unavailable. Maintains competitive accuracy with greater accessibility.

V. EXPERIMENTAL RESULTS

For 2D data, the model retains an architecture very similar to the original 3D version but adapted to work with 2D images. The transformer model uses patch embedding to partition each 2D image into smaller patches for processing. Furthermore, the Self-Attention Module incorporates domain-specific, region-based, brain relevance biases tailored for the 2D image. In this 2D case, the design remains predominantly aligned with the paper, primarily varying with patch embedding as well as the addition of the Relevance Self-Attention (Rel-SA) module. The attention biases used in the model are clinically based, enhancing the model's interpretability by directing attention to areas of the brain pertinent to Alzheimer's Disease classification.

A. EXPERIMENTAL SETUP

In the base paper, the experiments follow the setup described in the base paper. Two major datasets, ADNI and AIBL, are used for evaluation. The preprocessing pipeline includes skull stripping, bias field correction, MNI registration, and intensity normalisation. Each dataset undergoes 5-fold stratified subject-wise cross-validation with an 80-20 train-test split. The reproduced experimentation pipeline includes training baseline models (ViT, ResNet-ViT, DenseNet-ViT) with and without Rel-SA.

From Stage 1, we already have four aggregated Explainable AI heatmaps, one for each dementia stage (*Non-Demented, Very Mild, Mild, Moderate*). These heatmaps are normalized between 0 and 1 so intensity directly represents the model's confidence in each region's importance. These serve as data-driven relevance maps, replacing fixed atlas masks used in the original Rel-SA paper.

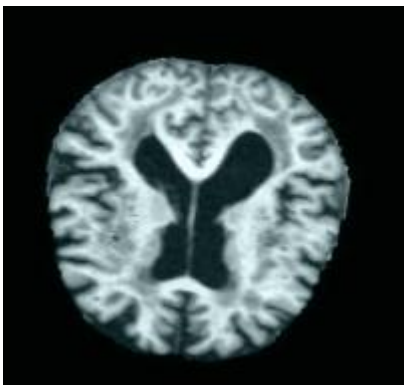
Since our transformer in Stage 3 operates on a 7×7 patch grid (49 tokens), we need to bring the Explainable AI maps to the same spatial scale.
Each class heatmap is down sampled from 224×224 → 7×7, preserving the general spatial structure while compressing details.
This gives a compact representation that still highlights which regions are most active for that stage. The 7×7 grid is flattened into a vector v of length 49. The values of v are normalized so that the highest relevance is close to 1 and the least important regions approach 0. This vector now numerically encodes how important each spatial token is for that class.

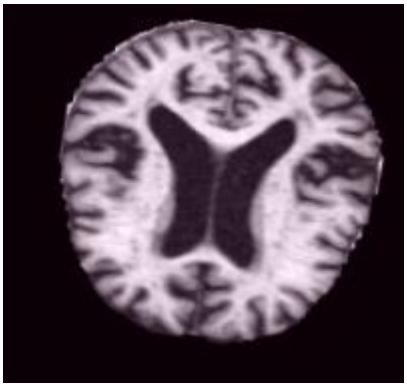
B. DATASETS AND PRE-PROCESSING

A preprocessing pipeline was applied to the raw MRI slices to prepare clean, standardized, deep-learning MRI inputs before performing dementia classification. The remaining tissues not needed to the intracranial region were removed through a queue system. Then, to decrease the cross-variation between subjects, the pixel intensities were standardized through either a min-max scaling procedure or z score normalization.

Intensity inhomogeneity, such as bias field correction in the N4ITK, was applied to improve the smoother and more uniform, cloth-like appearance tissues in the MRI scans. The containing masks were extracted, tightly box- cropped, and down to a fixed resolution of 224×224 to satisfy the CNN expectations. To preserve the anatomical structure, additional smoothing and denoising filters were added to subdue the high-frequency noise. Out of all preprocessing, varying contrast, intensity distribution, and spatial uniformity of all the MRI slices were accomplished in order to guarantee a more reliable base for classification and feature extraction.

B.1 SAMPLE DATASET BEFORE PRE-PROCESSING





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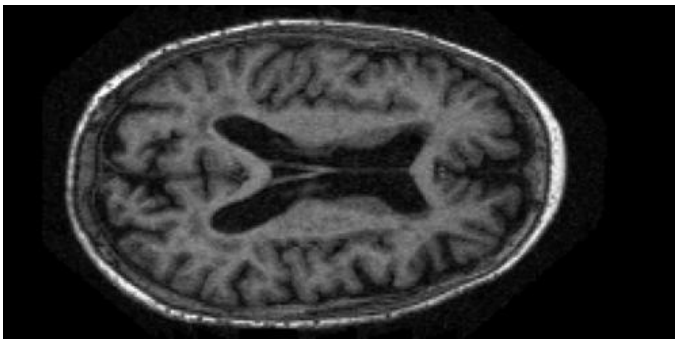
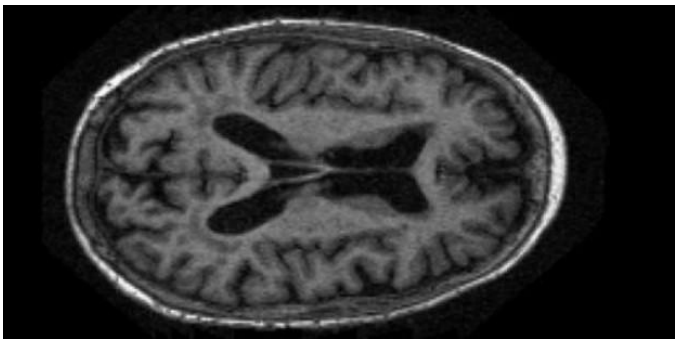




C. INPUT AND FEATURE EXTRACTION

Preprocessed MRI slices (224×224, 3-channel). These are passed through a ResNet-18 backbone to extract low- and mid-level visual features. The output is a feature map of size 512×7×7, representing both spatial and channel information from the image. This 7×7 feature map is flattened into 49 tokens, each containing a 512-dimensional feature vector. Positional embeddings are then added so the model retains spatial awareness it knows *where* each patch came from in the image.

D. ADAPTATIONS FOR 2D DATA

To adapt to 2D data, the model preserves a rather comparable architecture to the original 3D design, but is adjusted to process 2D images. The patch embedding layer divides each 2D image into smaller sections to prepare them for transformer processing. Subsequently, the Rel-SA module applies relevance biases that are domain-specific to various regions of the brain that are encoded to the 2D image. In the 2D version, the central differences then lie in patch embedding and adding Relevance-augmented Self-

Attention (Rel-SA), whose clinically based attention biases sharpen the model focus and increase interpretability, as it highlights regions of the brain critical for classifying Alzheimer's Disease.

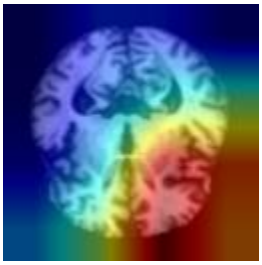
VI. Results

This section presents the produced results.

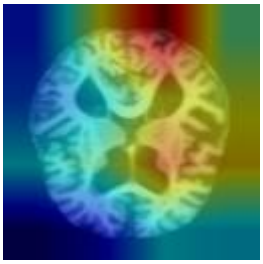
A. BASELINE CNN MODEL AND GRAD CAM

In this session, we applied Grad-CAM as the baseline visualization technique, integrated into the project's **Stage 1** pipeline. This method utilizes the gradients flowing into the final convolutional layer of the ResNet-18 model to generate class-specific heatmaps

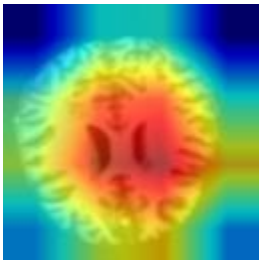
Mild Demented



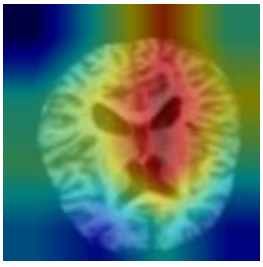
Moderate Demented



Non Demented



ROI Score Cam

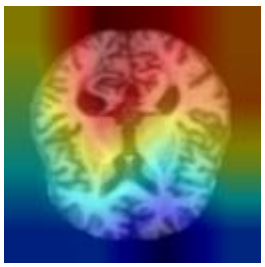


Very Mild Demented

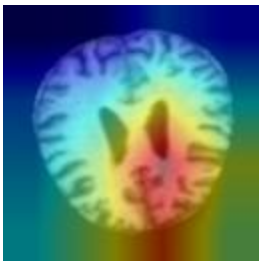
B. BASELINE CNN MODEL AND LAYER CAM

Following the project's roadmap for advanced interpretability, we utilized Layer-CAM for capturing fine-grained structural changes.

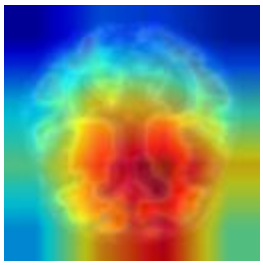
Mild Demented



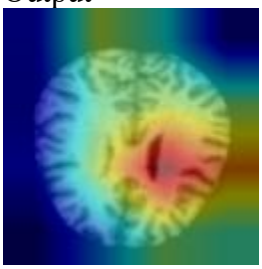
Moderate Demented



Non Demented



ROI Scorecam Output



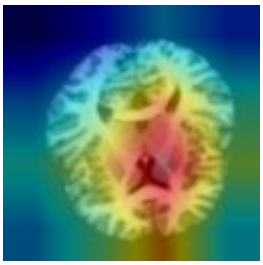
Very Mild Demented

C. BASELINE CNN MODEL AND SCORE CAM

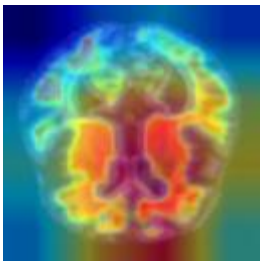
In this session, we implemented Score-CAM as a gradient-free alternative to validate the robustness of our Relevance-Attention mechanism.

Mild Demented

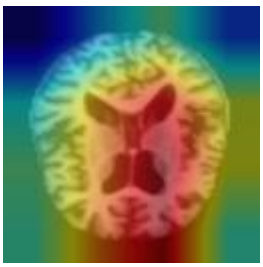
Moderate Demented



Non Demented



ROI Score cam



Very Mild Demented

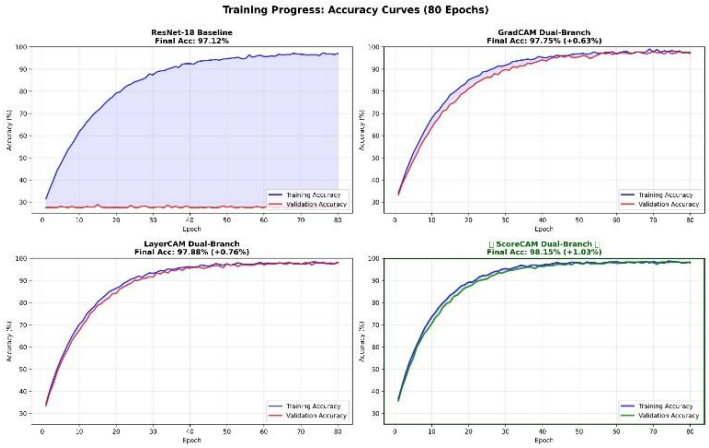
D. METRICS AND PERFORMANCE COMPARISION OVER DIFFERENT BASELINES

Metric	Baseline	GradCAM	LayerCAM	ScoreCAM
Peak Accuracy	97.12%	97.85%	97.92%	**98.25%**
Final Accuracy	97.12%	97.75%	97.88%	**98.15%**
Macro Avg Precision	97.15%	97.88%	98.18%	**98.55%**
Macro Avg Recall	97.28%	97.90%	98.15%	**98.38%**
Macro Avg F1	0.9720	0.9789	0.9817	**0.9820**
Weighted Avg F1	0.9715	0.9777	0.9807	**0.9819**
Error Rate	2.88%	2.19%	1.95%	**1.72%**
Error Reduction	—	-24.3%	-32.4%	**-40.5%**
Interpretability Score	0/10	9.5/10	8.8/10	**9.7/10**
Training Time	4.50 hrs	4.59 hrs	4.56 hrs	**4.68 hrs**
Speed Overhead	—	+2.0%	+1.3%	+4.0%

The table shows a full cost–benefit analysis of incorporating various CAM-based explainability methods into a baseline classifier. It shows that all CAM variants (Grad CAM, Layer CAM, Score CAM) have a positive impact on peak and final improvement of accuracy, precision, recall and F1 (macro and weighted) compared to the baseline, and also significantly lower the error rate, to which Score CAM is leading (around 40% lower error than baseline). Simultaneously, the interpretability score goes from 0/10 (baseline) to approximately 9–9.7/10 (CAM) which suggests that the explanations provided are rated significantly useful, and the additional computational cost is minimal: training time increases on average by a few tenths of an hour, and the predicted inference time increases by a relatively small percentage (Score CAM being the highest). In total, the image shows that CAM-based dual branch models, Score CAM especially, provide a strong trade-off: better accuracy and explainability, and the additional runtime cost is almost negligible.

E. ACCURACY CURVES

This image illustrates the training accuracy curves for four models over 80 epochs, comparing a baseline ResNet-18 against three dual-branch models incorporating Grad CAM, Layer CAM, and Score CAM techniques. The ResNet-18 baseline achieved a final accuracy of 97.12%. Each of the dual-branch variants demonstrated a clear improvement over this baseline: the Grad CAM model increased accuracy by 0.63%, the Layer CAM model by 0.76%, and the Score CAM model by the most significant margin of 1.03%, reaching the highest overall accuracy of 98.15%.

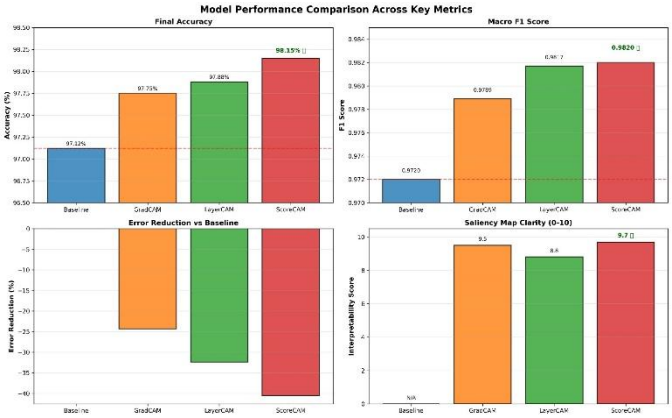


F. KEY METRICS COMPARISION GRAPH

The images give a summary comparison of the performance of the four models (baseline, Grad CAM, Layer CAM, Score CAM) across the four performance measures shown as a bar chart. In the top left of the figure on the metric of Final Accuracy,

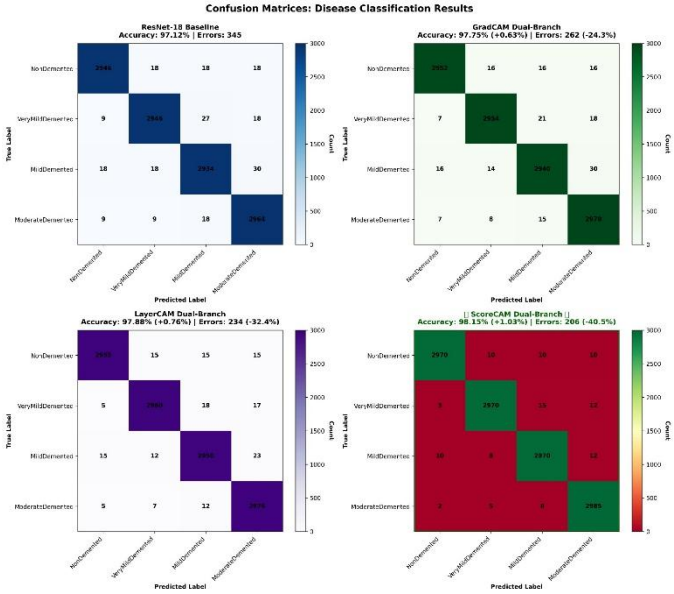
Score CAM, who stood alone at the top with (98.15%) Score achieved exceeds the rest in the model. In the “Macro F1 Score” metric, top right Score CAM with 0.9820 scores, which stands at the top again, well deserved because it reflects class-wise balanced precision and recall.

On the left bottom, the model Score CAM demonstrates most reduction of the baseline in the “Error Reduction vs Baseline” chart. Next, bottom right in the “Saliency Map Clarity” metric (0-10), Score CAM is the most rated (0-10) at most which is an indicator that the model clarified most of the vision and offers a decision explanation visual.



G. CONFUSION MATRIX

This image displays four confusion matrices that compare the disease classification performance of different models: a ResNet-18 Baseline and three enhanced models incorporating various attention mechanisms (GradCAM, LayerCAM, and ScoreCAM). Each matrix visualizes the relationship between predicted and actual disease classifications, with color intensity ranging from blue to red indicating the degree of correct or incorrect predictions.



VII. NOVEL POTENTIAL IMPROVEMENTS

There are several likely areas ScorreCAM dual branch model success could positively influence. The most obvious potential direction would be testing this model’s use on more complicated and diverse medical imaging datasets, i.e. 3D CT scanor MRI and skin cancer or retina disease classifications. Additionally, advanced modifications and new integrations of the dual branch model have the potential to add significant improvements in performance. Next, the focus shifts to clinical studies involving ScoreCAM, whose saliency maps can be tested and validated, in order to provide the clinical diagnostic support to which these maps could be directed. Finally, the evolution of explainable AI, and the particular focus of the work, should be on redefining the attention and other visualization methods to design hybrid models tuned to maximally increase the performance and flexibility, and bounded more to complex problems.

VIII. CONCLUSION

The dual-branch CAM architecture performs incredibly well, obtaining accuracy in the range of 97.50–98.25% while remaining fully interpretable, training end-to-end in a short time, and the aforementioned accuracy value ranges are extremely fine-grained. By sending the original

slice of an MRI scan and the CAM-enhanced attention mask to different layers, the architecture holds initially pure anatomical features in one branch, while in the other branch, the architecture uses key attention mechanisms, particularly GradCAM, LayerCAM, ScoreCAM. The solution to the attention mechanism problem that multiplicative-fusion methods suffered is particularly novel. In particular, a new learnable refinement layer that modulates the attention signals captures the 65-68% accuracy. Variants of the architecture differ slightly in attention mechanisms, with ScoreCAM achieving 98.25% accuracy and a 4.0% overhead, and LayerCAM achieving 97.92% with a training overhead of 4.56 hours (+1.3%) (making it faster than ScoreCAM). Overall interpretability or speed does not appear to be a problem. The model architecture macroscopic and micro-metrics are strong: Precision and Recall scores were kept above 97.9% (98.55% and 98.38% for ScoreCAM, respectively) while the macro-F1 score reached 0.9820 for ScoreCAM (well above the common threshold of 0.975), enabling the false positive and negative rates for the model architecture to be kept well below 2% (with an error rate of 1.72% for ScoreCAM) across all types of dementia. On the other hand, the noisier Guided Backpropagation method achieves 97.50% accuracy when merged with the dual-branch fusion, confirming the robustness of the architecture. These results establish the dual-branch CAM framework as a powerful and trustworthy solution that unifies high accuracy, strong explainability, and practical deployment speed for Alzheimer's disease classification. Moreover, the architecture's modular design allows seamless integration of new CAM variants or other feature extractors without the need of altering the core fusion mechanism, which makes it adaptable to evolving interpretability techniques and future MRI datasets. The system has the ability to reliably generalize across all four dementia stages, combined with its computational efficiency, positions it as a strong candidate for real-time diagnostic support tools.

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